Aadhav Sivakumar

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Education

New York University, Tandon School of Engineering — Brooklyn, NY
Master of Science in Mechatronics and Robotics | GPA 3.95 — *Aug 2024 – May 2026*
- **Relevant Coursework** | Reinforcement Learning & Optimal Control (MPC, iLQR/DDP, Q-learning, actor-critic)

University of California, Santa Cruz — Santa Cruz, CA
Bachelor of Science in Robotics Engineering, Minor in Electrical Engineering | GPA 3.8 — *Jun 2020 – Jun 2024*

Professional Experience

AI Engineer — Jan 2026 – Sep 2026
Roboflow | Edge AI Engineering Team — *New York, NY*
- Evaluated NVIDIA's Cosmos 3 omnimodal world model family and benchmarked Cosmos 3 Edge on Jetson Thor.
- Worked with NVIDIA to apply synthetic data generation with PAIDF AnomalyGen to improve vision model performance from few real defect examples; found training iterations, not dataset size, drive the gains.
- Trained and deployed computer vision models for detection, segmentation, and keypoints (RF-DETR, YOLO) with neural architecture search, dataset augmentation, and TensorRT/ONNX inference on multi-camera Jetsons.
- Technical lead (DRI) for the Roboflow x Standard Bots partnership: built the active-learning loop feeding robot-captured imagery back into training sets and root-caused offline inference failures on robots.
- Designed a custom Docker container for robot control, driving an arm directly from user-configurable Roboflow Workflows as a sidecar to the inference stack; demoed as closed-loop visual servoing on an xArm 5.

Graduate Robotics Researcher — Jan 2026 – Jun 2026
LAIR, NYU Tandon School of Engineering | Advised by Professor Christopher Clark — *Brooklyn, NY*
- Developed an Isaac Sim Real2Sim pipeline turning 3D Gaussian Splatting scans into digital twins, generating 1k+ randomized scripted demonstrations at 100% in-sim success for sim-to-real augmentation.
- Fine-tuned π_0 and $\pi_{0.5}$ flow-matching vision-language-action (VLA) policies (openpi/JAX) on bimanual Ultra robot teleoperation data, benchmarked against an ACT baseline.
- Built a bimanual SO-ARM101 LeRobot teleoperation and data collection stack on a Jetson Orin Nano.
- Assembled and deployed a BlueROV2 for 3D reconstruction of underwater shipwrecks.

Graduate Teaching Assistant — Sep 2025 – May 2026
NYU Tandon School of Engineering | Foundations of Robotics, Math for Robotics, Mechatronics — *Brooklyn, NY*
- Built a Python UR10e MuJoCo simulation as the final project to assess forward/inverse kinematics and dynamics.
- Taught Jacobian-based IK, state-space control, and motion planning to 80+ graduate students in weekly sessions.
- Wrote the Mechatronics midterm and STM32 NUCLEO-H503RB reference code (USART, ADC, servos, sensors).

Field Robot Technician — Aug 2025 – Jan 2026
Starship Technologies | Fordham University Hub — *Bronx, NY*
- Root-cause analysis on fleet faults; repaired boards, sensors, and power systems; camera/LiDAR calibration.

Undergraduate Research Assistant — Jun 2023 – Jun 2024
UCSC Tactile Manipulation Lab | Professor Tae Myung Huh — *Santa Cruz, CA*
- Developed soft robotic CapSense tactile sensors on flexible PCBs, enabling slip-based reorientation at 78% success.

Projects

Real2Sim2Real SO-ARM101 | *Isaac Sim, Isaac Lab, gsplat, COLMAP, LeRobot, ACT, Jetson* — Sep 2026 – Oct 2026
- Reconstructed the scene with COLMAP and Gaussian splatting, calibrated it to the real cameras in Isaac Sim, and generated about 4k randomized Isaac Lab demos with domain randomization and DAgger-style recovery.
- Co-trained an ACT imitation learning policy on real and sim data: 19/21 real-world successes vs. 6/21 real-only.

Unitree Go2 RL Locomotion | *Isaac Lab, RSL-RL, Slurm on NYU HPC* — Nov 2025 – Dec 2025
- Trained a PPO locomotion policy with a custom PD controller, Raibert gait heuristic, and reward shaping.

Glass-2-Bot Mobile Manipulator | *Google Glass, Raspberry Pi, Arduino, OpenCV* — Apr 2025 – May 2025
- Object detection on Google Glass video streamed to a Raspberry Pi drives an Arduino grasping state machine.

Technical Skills

Languages & Tools | Python, C++, C, CUDA, Bash, MATLAB, Linux, Git, Docker, Slurm, ROS 2, xArm SDK
Robot Learning & Embodied AI | VLAs, imitation learning, behavior cloning, RL (PPO), DAgger, teleoperation
Deep Learning | PyTorch, JAX, Hugging Face LeRobot, openpi, pi0/pi0.5, SmolVLA, GR00T, ACT, Cosmos
Simulation & 3D | Isaac Sim, Isaac Lab, MuJoCo, Gaussian Splatting, COLMAP, domain randomization, sim-to-real
Perception & Deployment | machine learning, computer vision, synthetic data, sim/real co-training, Jetson, TensorRT